

Conghao Li

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EDUCATION

B.E. in Internet of Things Engineering

Sep 2023 – Jun 2027 (expected)

Beijing University of Posts and Telecommunications (BUPT), International School

Dual-degree program with Queen Mary University of London (QMUL)

- GPA: 3.7/4.0
- Selected Coursework: Advanced Mathematics I (96), Discrete Mathematics (92), Data Structures (93), Introduction to Electronic System (94), Digital Circuit Design (92), Machine Learning (90)
- IELTS: 6.5 (L 8.0 / R 6.5 / W 6.0 / S 5.5); CET-6: 532

RESEARCH EXPERIENCE

Undergraduate Researcher,

Oct 2024 – Present

Key Laboratory of Universal Wireless Communications (Ministry of Education) — BUPT

Advisors: Gang Chuai (Prof.), Weidong Gao (Assoc. Prof.), Kaisa Zhang (Assoc. Researcher)

- Conducted system-level modeling and performance evaluation for non-terrestrial networks (NTN) and multi-layer LEO constellations under differentiated regional traffic demands, focusing on trade-offs among coverage, connectivity, robustness, capacity, and cost.
- Built an STK–MATLAB co-simulation workflow for orbital visibility and coverage analysis, and evaluated dynamic GSL/ISL availability, connectivity, and survivability under time-varying satellite topologies.
- Contributed to a TVT paper on multi-layer LEO constellation design and an EI conference paper on dynamic user clustering and group handover in dense LEO scenarios, supporting modeling, evaluation, experiments, and result analysis.
- Co-authored an IEEE GLOBECOM 2026 submission on graph-informed dynamic inter-layer link allocation for multi-layer LEO satellite networks, focusing on figure design, manuscript writing, and result presentation.
- First author of a Drones paper on QoS-aware downlink paging control for UAV-assisted 5G-Advanced networks, leading problem formulation, methodology design, simulation development, performance evaluation, and manuscript preparation.
- Participated in three funded research projects at the national, municipal, and industry levels.

Undergraduate Researcher,

Sep 2025 – Present

State Key Laboratory of Networking And Switching Technology— BUPT

Advisors: Wenjia Xu (Assoc. Prof.), Mugen Peng (Prof.)

- Participated in a hyperspectral–LiDAR fusion study for land-cover classification, focusing on LiDAR-guided channel-wise adaptation and cross-modal feature fusion.
- Supported module prototyping, comparative experiments, result analysis, and manuscript preparation for an IEEE IGARSS 2026 submission.

SELECTED PROJECTS

Project Lead,

Sep 2025 – Present

“Xingdian Xunlian ” — Edge-Computing-Enabled Multi-Orbit Satellite Collaborative Monitoring System —

National College Innovation Project (Undergraduate)

Advisors: Weidong Gao (Assoc. Prof.), Kaisa Zhang (Assoc. Researcher), Pei Qie (Engineer)

- Led a national undergraduate innovation project on edge-computing-enabled multi-orbit satellite collaborative monitoring for remote power-line inspection and environmental monitoring.
- Integrated edge AI and GEO/LEO hybrid satellite communication for remote-area monitoring, connectivity, and on-site intelligence.
- Coordinated system design, technical roadmap, project materials, and team deliverables with advisors and State Grid stakeholders.

PUBLICATIONS

- **Conghao Li**, Haizhi Yu, Weidong Gao, Dengyan Wang, Shouhui Lai, Xu Zhao, Hongzhi Zhang, and Gengshuo Liu. “QoS-Aware Downlink Paging Control for UAV-Assisted 5G-Advanced Networks with On-Demand Coverage.” *Drones*, 2026, 10(3): 191. [doi: 10.3390/drones10030191](https://doi.org/10.3390/drones10030191). IF: 4.8; JCR Q1 (Remote Sensing)
- **Conghao Li**, Weidong Gao, Weijian Guo, Kaisa Zhang, Xiangyu Chen, and Haizhi Yu. “A Multi-Feature Fusion Approach for Dynamic User Clustering in LEO Satellite Group Handover.” The 5th International Conference on Frontiers of Electronics, Information and Computation Technologies (ICFEICT 2025), [doi: https://doi.org/10.1007/978-981-95-8660-8_35](https://doi.org/10.1007/978-981-95-8660-8_35)
- Haizhi Yu, **Conghao Li**, Weidong Gao, Kaisa Zhang, Huan Zhang, and Jin Shao. “Optimized Design of Multi-Layer LEO Satellite Constellation for Regional Traffic Demand in Non-Terrestrial Networks.” *IEEE Transactions on Vehicular Technology* (TVT) , [doi: 10.1109/TVT.2026.3670306](https://doi.org/10.1109/TVT.2026.3670306). IF: 7.1; JCR Q1
- Siyuan Wan, Mugen Peng, Hongyu Wang, Yuanben Zhang, **Conghao Li**, and Wenjia Xu. “HLFN: A Hyperspectral-LiDAR Fusion Network with LiDAR-guided Channel-wise Adaptation.” *IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*, Accepted.
- Haizhi Yu, Kaisa Zhang, **Conghao Li**, Weidong Gao, Saidiwaerdi Maimaiti, and Qingrui Guo. “Graph-Informed Dynamic Inter-Layer Link Allocation for Multilayer LEO Satellite Networks.” *IEEE Global Communications Conference (GLOBECOM 2026)*, under review.

HONORS & AWARDS

- **Honorable Mention** (H Award), Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling (COMAP), Jan 2025.
- **Third Prize**, National College Students Mathematics Competition (Beijing Division), Nov 2025.
- **University-level Scholarship**, Beijing University of Posts and Telecommunications.

SKILLS

- **Programming**: Python, C, Java; Web: HTML; Scientific Computing: MATLAB; Embedded Development: C-based microcontroller programming with Keil MDK, GPIO/ADC/UART configuration
- **Tools/Platforms**: STK, LaTeX, Git, Keil MDK; Familiar with Linux-based development workflows, serial communication, and basic hardware debugging
- **Research Methods**: satellite orbit/link modeling, clustering (K-means/spectral methods), multi-objective optimization (evolutionary algorithms), system-level simulation & evaluation